Project Plan

1 Introduction

Day-ahead electricity price forecasting has become harder over the past decade. The expansion of wind and solar generation introduces supply-side volatility that statistical models struggle to capture, and the coupling of European bidding zones introduce price shocks across borders in ways historical data may not reflect. Accurate forecasts remain essential for generators, retailers, and grid operators, but the methods used to produce and evaluate those forecasts have not kept pace.

Lago et al. [1] documented how severe the evaluation problem is. New methods are typically compared on private datasets, over test periods too short to be conclusive, and against baselines too weak to be informative. Training splits, model inputs, and datasets are frequently left unspecified, making replication difficult. To address this, Lago et al. proposed an open-access benchmark consisting of five historical datasets, two baseline models, and a Python toolbox for standardized evaluation. One of the baselines is the Lasso Estimated AutoRegressive model (LEAR), which uses the LASSO estimator for joint feature selection and coefficient estimation over a parameter-rich input space. Despite its simplicity, LEAR matched or outperformed deep learning alternatives when evaluation was conducted rigorously over long test periods and across multiple markets. This model and benchmark form the forecasting foundation for this thesis.

Yet even the most thorough contributions to EPF stop at the boundary of offline evaluation. Weron [2] surveyed decades of forecasting research across statistical, reduced-form, fundamental, and computational intelligence approaches. It remains the most comprehensive review the field has produced, and it contains no discussion of deployment, scheduling, or monitoring. An updated survey by Nowotarski and Weron [3] emphasized probabilistic methods but remained limited to modeling techniques. The toolbox of Lago et al. [1] standardized how models are compared but not how they are run. The pattern is consistent. The EPF literature treats forecasting as a problem that ends when predictions are done. Ingesting data on a schedule, versioning experiments, detecting model degradation, and retraining automatically are not addressed by any of these works.

The gap between building a model and running it in production is not specific to electricity forecasting. The field of Machine Learning Operations (MLOps) emerged to close exactly this gap by applying DevOps principles to ML workflows. Stone et al. [4] reported that as of 2020, 55 percent of organizations actively using ML had not deployed a single model, and 18 percent needed more than 90 days to do so. Zarour et al. [5] synthesized the MLOps literature into a systematic review identifying nine core practices. These are CI/CD automation, workflow orchestration, reproducibility, versioning, collaboration, continuous training and evaluation, metadata tracking, continuous monitoring, and feedback loops. The review also proposed a maturity model with five levels, ranging from fully manual development at Level 0 to fully automated platforms at Level 4.

This thesis takes the LEAR benchmark model from the EPF literature and embeds it in an end-to-end pipeline designed along MLOps principles. The pipeline covers the full lifecycle of a forecasting system. Data is ingested from the ENTSO-E Transparency Platform on a recurring schedule and stored in ClickHouse for structured analytical access. Forecasts are produced through rolling-window calibration with periodic refit-

ting, executed inside Docker containers and orchestrated by Prefect. A monitoring layer built on Prometheus and Grafana tracks infrastructure health, forecast accuracy over time, and shifts in the underlying data distribution. When a decrease in performance is detected, the system triggers retraining automatically rather than waiting for manual intervention. The system is evaluated on two axes, the forecast quality is measured by statistical metrics while the operational maturity is evaluated against the nine practices and the maturity model proposed by Zarour et al. [5]. Table 1 maps each practice to the specific technology and pipeline problem it addresses.

1.1 Research Questions

1. How does the LEAR model perform on recent ENTSO-E day-ahead price data for the German bidding zone, and how sensitive is forecast accuracy to the length of the rolling calibration window?
2. What engineering trade-offs and operational constraints emerge when embedding an EPF benchmark model into a scheduled, containerized production pipeline?
3. Which of the nine MLOps practices identified by Zarour et al. [5] prove most critical for gaining maturity levels, and where does the implemented system reach its limit?

1.2 Project Details and Solution Approach

Table 1: Proposed Open Source Tech Stack by MLOps Practice

ID	Practice	Technology	Pipeline Problem Solved
P1	Automating CI/CD	GitHub Actions, pytest, Ruff	Automated testing, linting, and build validation on every commit.
P2	Orchestrating workflows	Prefect	Scheduled data ingestion and forecast flows with retries and task dependencies.
P3	Ensuring reproducibility	Docker, Docker Compose	Containerized execution ensures identical environments across development and production.
P4	Versioning	Git, ClickHouse	Code versioned in Git. Data versioned through append-only storage with ingestion timestamps.
P5	Collaboration	GitHub, uv workspaces	Monorepo with multiple independent packages enabling parallel development.
P6	Continuous Training	Prefect	Rolling-window calibration with configurable refit intervals on new data.
P7	Metadata tracking	Prefect, WAND	Each run produces versioned forecast CSVs and metric files.
P8	Continuous monitoring	Prometheus, Grafana, WAND	Infrastructure health checks, model dashboards, and data drift detection.
P9	Feedback loops	Prometheus alerts, Prefect	Drift or error threshold breach triggers automatic retraining.

1.3 Project Board

The project is managed using GitHub Projects, which provides a task board integrated directly with the code repository. Each task is represented as an issue and moves through stages from planned to completed, giving a clear overview of progress at any point. Figure 1 shows the current state of the board.

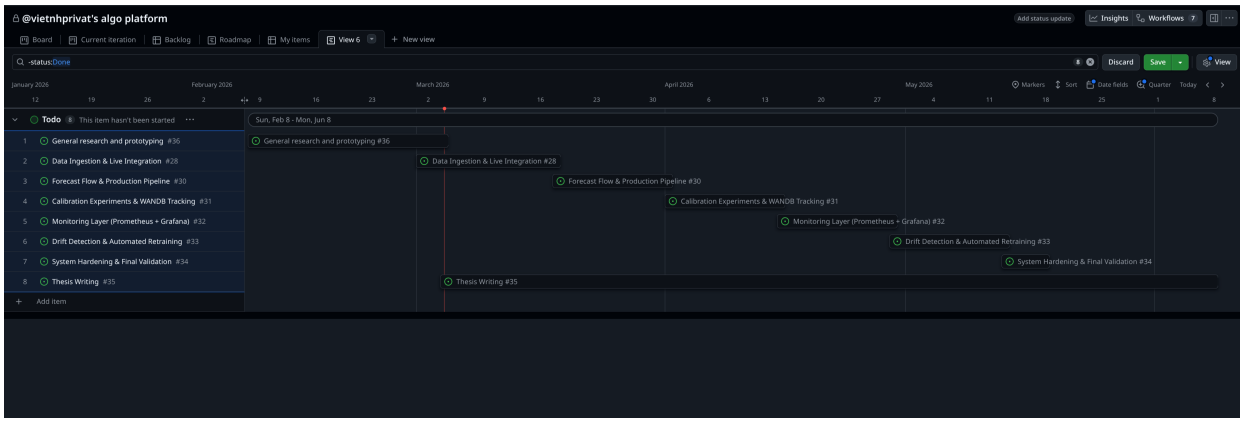


Figure 1: Project Board

References

[1] Lago, J., Marcjasz, G., De Schutter, B., & Weron, R. (2021). Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark.

[2] Weron, R. (2014). Electricity price forecasting: A review of the state-of-the-art with a look into the future.

[3] Nowotarski, J., & Weron, R. (2018). Recent advances in electricity price forecasting: A review of probabilistic forecasting.

[4] Stone, R., et al. (2025). MLOps: A review of the state of the field.

[5] Zarour, M., et al. (2025). A systematic review of MLOps best practices.